

# Implementation of File Handling: Reading and Writing Files and Finding the Frequency of Patterns

Experiment No: 10 (Week 10)      Date: \_\_\_\_ / \_\_\_\_ / 2026

**Title:** File Handling: Writing to sample.txt and Finding the Frequency Count of the Patterns 'pe' and 'pi'

## ■ Objective / Aim

To write the given text into a file named `sample.txt`, read the file back using Java I/O classes, and find the frequency count of the patterns 'pe' and 'pi' in the file content.

## ■ Mapped Outcomes

**Course Outcome (CO):** CO2

**Program Outcomes (POs):** \_\_\_\_\_

## ■ Problem Statement

Write the following text in the file called `sample.txt` and find the frequency count of the patterns 'pe' and 'pi':

```
Peter Piper picked a peck of pickled peppers
A peck of pickled peppers Peter Piper picked
If Peter Piper picked a peck of pickled peppers
Where's the peck of pickled peppers Peter Piper picked?
```

**Expected Output:** 'pe' - no of occurrences - 20    |    'pi' - no of occurrences - 12

## ■ System Requirements

### Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

### Software

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

## ■ Algorithm / Logic Description

1. Start the program.
2. Import `FileWriter`, `FileReader`, `BufferedReader` and `IOException` from the `java.io` package.
3. Store the four lines of text in a `String` array and create the file `sample.txt` using `FileWriter`.

4. Write each line into the file followed by a line separator, then close the file (try-with-resources).
5. Open the file using `BufferedReader` wrapped around `FileReader`.
6. Read the file line by line using `readLine()` until it returns null and append each line to a `StringBuilder`.
7. Convert the content to lower case so that the search is case-insensitive.
8. Define `countPattern(text, pattern)`: set `count = 0` and find the first index using `indexOf()`; while the index is not -1, increment count and search again from `index + 1`.
9. Call `countPattern` for the patterns 'pe' and 'pi'.
10. Display the file content and the frequency count of each pattern.
11. Handle `IOException` and terminate the program.

**Note:** The count is case-insensitive ('Peter' and 'Piper' are counted along with 'peck', 'pickled' and 'peppers'). A case-sensitive search would give 16 and 8, which does not match the expected output.

## ■ Source Code

### Folder Structure

```
week10/  
|-- FileFrequency.java  
`-- sample.txt    (created when the program runs)
```

Default package    Class: **FileFrequency**    (File: FileFrequency.java)

```

import java.io.BufferedReader;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

public class FileFrequency {

    // Counts occurrences of a pattern in the text (case-insensitive)
    static int countPattern(String text, String pattern) {
        int count = 0;
        int index = text.indexOf(pattern);

        while (index != -1) {
            count++;
            index = text.indexOf(pattern, index + 1);
        }
        return count;
    }

    public static void main(String[] args) {
        String fileName = "sample.txt";

        String[] lines = {
            "Peter Piper picked a peck of pickled peppers",
            "A peck of pickled peppers Peter Piper picked",
            "If Peter Piper picked a peck of pickled peppers",
            "Where's the peck of pickled peppers Peter Piper picked?"
        };

        // Step 1: Write the text into sample.txt
        try (FileWriter fw = new FileWriter(fileName)) {
            for (String line : lines) {
                fw.write(line + System.lineSeparator());
            }
            System.out.println("Text written to " + fileName);
        } catch (IOException e) {
            System.out.println("Error writing file: " + e.getMessage());
            return;
        }

        // Step 2: Read the file and build the full text
        StringBuilder content = new StringBuilder();
        try (BufferedReader br = new BufferedReader(new FileReader(fileName))) {
            String line;
            while ((line = br.readLine()) != null) {
                content.append(line).append("\n");
            }
        } catch (IOException e) {
            System.out.println("Error reading file: " + e.getMessage());
            return;
        }

        // Step 3: Count the patterns (ignoring case)
        String text = content.toString().toLowerCase();
        int peCount = countPattern(text, "pe");
        int piCount = countPattern(text, "pi");

        System.out.println("\nFile Content:");
        System.out.print(content);

        System.out.println("\n'pe' - no of occurrences - " + peCount);
        System.out.println("\n'pi' - no of occurrences - " + piCount);
    }
}

```

## ■ Compilation & Execution Commands

Save the source code exactly as `FileFrequency.java` inside a folder named `week10`, then open a terminal in `week10/`:

Compile the program:

```
javac FileFrequency.java
```

Run the compiled program:

```
java FileFrequency
```

## ■ Test Cases & Execution Data

| Test Case           | Input                 | Expected Output                 | Status |
|---------------------|-----------------------|---------------------------------|--------|
| TC-01: Create file  | Four lines of text    | sample.txt created with 4 lines | Passed |
| TC-02: Pattern 'pe' | pe (case-insensitive) | 'pe' - no of occurrences - 20   | Passed |
| TC-03: Pattern 'pi' | pi (case-insensitive) | 'pi' - no of occurrences - 12   | Passed |

## ■ Output Screen

The program was compiled and executed from the terminal. The console output is shown below.

### Example

```
Text written to sample.txt
```

```
File Content:
```

```
Peter Piper picked a peck of pickled peppers  
A peck of pickled peppers Peter Piper picked  
If Peter Piper picked a peck of pickled peppers  
Where's the peck of pickled peppers Peter Piper picked?
```

```
'pe' - no of occurrences - 20
```

```
'pi' - no of occurrences - 12
```

Content of `sample.txt` created by the program:

```
Peter Piper picked a peck of pickled peppers  
A peck of pickled peppers Peter Piper picked  
If Peter Piper picked a peck of pickled peppers  
Where's the peck of pickled peppers Peter Piper picked?
```

## ■ Learning Outcome / Result

Successfully implemented and verified the file handling program. The text was written into `sample.txt` using `FileWriter`, read back using `BufferedReader`, and the frequency of the patterns was found as 20 for 'pe' and 12 for 'pi'. The experiment demonstrated reading and writing files, try-with-resources, exception handling with `IOException` and string pattern searching in Java.